# Machine-Learning Surrogates for MaxActive Channel Selection in Terahertz Drone-to-Drone Communications

Korhan Erdoğdu, Özgür Gürbüz, Akhtar Saeed, and Mikail Erdem

***Abstract*—Terahertz (THz) drone-to-drone links promise multi-gigabit capacity, but channel quality is highly sensitive to distance, molecular absorption, and mobility-induced beam misalignment. The MaxActive algorithm returns the capacity-maximizing channel subset, but it is search-based and must be re-executed at every geometry update, so its per-update cost becomes the bottleneck for an airborne platform. This correspondence asks whether machine learning can supply a low-complexity surrogate. From a physics-consistent THz model we reproduce the MaxActive capacity for four mobility patterns and benchmark eighteen regressors under a leakage-safe nested cross-validation protocol. A Gaussian process tracks the reference most closely, whereas single decision trees, plain boosting, and unregularized linear regression overshoot the physical bound. A dueling double deep $Q$-network (D3QN), formulated as a single-step contextual bandit rather than a sequential reinforcement-learning problem, learns the *selection* itself and therefore cannot exceed the reference, at the cost of per-context training. The learned decision is about two orders of magnitude cheaper than the search it replaces and is stored in a few kilobytes.**

***Index Terms*—Beam misalignment, channel selection, complexity, contextual bandit, cross-validation, deep reinforcement learning, drone communications, Gaussian process, machine learning, surrogate model, terahertz.**

## I. Introduction: Problem and Proposed Solution

TERAHERTZ (THz) bands (0.1–10 THz) offer the enormous contiguous bandwidth required by future aerial backbones and dense drone swarms, where fiber is impractical and lower bands are congested. Propagation in this regime, however, is uniquely hostile: it is dominated by frequency-selective molecular absorption, whose spectral lines carve the usable band into narrow transmission windows, and by a spreading loss that grows with the square of both distance and frequency [1]. The useful capacity of a drone-to-drone link therefore depends on a tightly coupled set of physical and system quantities: the three-dimensional distance, the antenna gain and beamwidth, the beam alignment, the per-channel molecular absorption and thermal re-radiation noise, the selected channel subset, and the power-allocation policy. Aerial mobility continuously perturbs the geometry, so beams that are aligned at one instant become misaligned at the next, collapsing the achievable capacity by many orders of magnitude, and making the online choice of which channels to activate a first-order determinant of link performance.

The reference solution to the spectrum-selection part of this problem is *MaxActive*, a threshold-sweep algorithm that returns the channel subset maximizing capacity for the current channel state. MaxActive is accurate and is treated throughout this work as the ground-truth reference, but it is search-based: it repeatedly evaluates capacity while sweeping a signal-to-noise ratio (SNR) threshold, and it must be re-executed whenever the geometry changes.

*Why a surrogate is needed.* That search cost is not incidental. The sweep re-evaluates capacity across every channel of a wide THz band, so a single selection dominates the per-update budget. Aerial mobility perturbs the geometry continuously, so the search must run at the geometry-update rate, on an embedded processor that is simultaneously flying the aircraft. A learned model returns the same decision at a small fraction of that cost and moves the training burden offline; Section VIII reports the complexity and the measurements behind this claim. The research question is therefore whether a learned *surrogate* can reproduce the MaxActive decision directly from the channel state at a fraction of the cost.

We pursue two complementary answers: *supervised regressors* that predict the MaxActive capacity value from a compact set of physical features, and a *dueling double deep Q-network* (D3QN) that learns the *selection decision* itself and evaluates the true capacity of its choice, so that it respects the reference bound by construction. The contributions are fourfold. (i) A physics-consistent reproduction of the MaxActive water-filling (WF) and equal-power (EP) capacity curves for *four* mobility patterns, including mobility over elevation and measured windy-hover acceleration traces. All are regenerated from the reference model rather than digitized from published figures, which gives a reusable ground truth for learning. (ii) The first systematic benchmark, to our knowledge, of eighteen regressors spanning six model families as surrogates for THz channel selection, under a leakage-safe nested cross-validation protocol suited to this small-sample, wide-dynamic-range regime. (iii) The contrast between learning the capacity *value* and learning the selection *decision*; the latter closely matches MaxActive and cannot violate the physical bound, at the cost of per-context training discussed in Section IV-C. (iv) A deployment-oriented complexity analysis that combines asymptotic cost with *measured* execution time and memory footprint. These

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four threads — ground truth, surrogate benchmark, value-versus-decision, and deployability — are followed throughout.

## II. Proposed ML Models versus MaxActive

Two facts organize the entire study. First, MaxActive returns the capacity-maximizing channel subset within its threshold-search family (Section IV-B), so within the adopted system model it upper bounds any surrogate trained to imitate it. When predictions are compared against that reference we therefore evaluate them at the ceiling,

$$\hat{C}_{\text{clip}} = \min\big(\hat{C},\, C_{\text{MaxActive}}\big). \tag{1}$$

Equation (1) requires $C_{\text{MaxActive}}$ — precisely what a deployed surrogate does not have, for if it were available online there would be no reason to learn a surrogate at all. *The clip is therefore an offline evaluation and visualization device only*. It is applied when plotting against the known reference, so that a displayed curve never claims an unattainable capacity. It is *not* part of the online prediction path: at deployment the surrogate emits $\hat{C}$ (or, for the D3QN, a selection) with no access to the reference. Accordingly, every quantitative error in Section VII is computed from raw, unclipped predictions, and the ceiling violations the clip would hide are reported explicitly in Table III. Second, the capacity spans roughly nine orders of magnitude across the scenarios (Table II), so the models are trained on $\log_{10}$ capacity, the natural scale on which a THz capacity curve is smooth.

The models fall into two categories. Supervised regressors reproduce the capacity *value*: cheap at inference, but able to exceed the reference bound, which is what makes unclipped reporting necessary. The D3QN reproduces the selection *decision*; since it evaluates the real capacity of the channels it selects, it is bounded above by MaxActive automatically and needs no clip at all. Fig. 1 overlays one representative of each model family, together with the D3QN, on the MaxActive curves for the aligned and diagonal scenarios; Table III reports all eighteen regressors.

## III. Alternative ML Solutions

To avoid committing to a single hypothesis class a priori, we evaluate eighteen regressors spanning six families, all named in Table III: linear/regularized (LinearReg, Ridge, Lasso, ElasticNet, BayesianRidge), robust linear (Huber), kernel/probabilistic (SVR-RBF, KernelRidge, GaussProc), instance-based ($k$-NN), neural (MLP), and tree ensembles (RandomForest, ExtraTrees, Bagging, AdaBoost, GradBoost, XGBoost) with a single DecisionTree as contrast. The rationale is physical. In the $\log$ domain the capacity is approximately affine in log-distance and gain, so the regularized-linear models are the natural baselines. The kernel and probabilistic models (SVR-RBF, kernel ridge, and a Gaussian process with a squared-exponential-plus-noise kernel [5]) capture smooth nonlinear structure, $k$-NN interpolates locally, and the MLP is a flexible approximator. The tree ensembles are strong tabular learners, but their piecewise-constant form cannot extrapolate the wide dynamic range. All models use scikit-learn [7] except XGBoost [6].

*Why these eighteen are enough.* The set is representative rather than exhaustive: it covers the hypothesis classes commonly used for small-sample tabular regression and, within each, spans the inductive biases that could plausibly differ on this target. The linear family appears at all four regularization regimes (none, $\ell_2$, $\ell_1$, elastic net) plus an evidence-based variant, which separates bias due to the affine hypothesis from bias due to the penalty, and a robust (Huber) variant isolates sensitivity to the extreme dynamic range. The kernel family appears in deterministic and probabilistic forms, and the tree family spans bagging, randomized splitting, and sequential boosting, with a single DecisionTree as the unregularized contrast. Models within a family behave alike here while families differ by orders of magnitude in error (Table III), so adding further members of an existing family is unlikely to alter the ranking, whereas omitting a family could. One further candidate, histogram gradient boosting, was implemented and discarded: designed for large-$n$ binned data, it degenerates on 33 samples (negative cross-validated $R^2$). The benchmark is therefore diagnostic, supporting a recommendation at the level of families rather than a single tuned result.

## IV. Background: System Model and Algorithms

### A. Key Expressions of the THz Link

This subsection introduces the link model and fixes the notation used in the rest of the paper; Table I gathers, in one place, every symbol and closed-form expression referred to later, so that the algorithm listing, the learning formulation, and the complexity analysis can be read without re-deriving the physics.

Let $f$ denote frequency, $d^t$ the Tx–Rx distance, and $\Delta f_i$ the bandwidth of channel $i$. The path loss combines free-space spreading and molecular absorption; the coefficient $k(f)$ comes from line-by-line (LBLRTM) transmittance data, with transmittance $\tau(f,d) = e^{-k(f)d}$. The spherical-lobe antenna gain is $4\pi/\theta_{bw}^2$ within the half-power beam of width $\theta_{bw}$ and a side-lobe level $\rho$ outside. In Table I, $A_i$ is the linear total path loss, $P_{N,i}$ the molecular absorption noise, $G = G_{\text{Tx}}G_{\text{Rx}}$ the total link gain, $P_T$ the total transmit power, $\Omega$ the active channel set, and $\mu$ the water level fixed by $\sum_i P_i^{\text{Tx}} = P_T$. EP splits $P_T$ evenly across the selected channels, whereas WF favors low-attenuation channels.

### B. MaxActive Channel Selection

MaxActive selects the active channel set by sweeping an SNR threshold and keeping, at each step, the channels whose SNR exceeds it; the subset of largest capacity is returned. Algorithm 1 states the procedure using the notation of Table I, where $\gamma^t(\cdot)$ denotes the SNR at time step $t$ and $Q$ the number of threshold steps of size $\Delta s$.

*In what sense MaxActive is optimal.* The search returns the best subset over all $Q$ thresholds, hence the best member of the family of *SNR-ordered nested subsets* $\Omega_1 \subset \Omega_2 \subset \ldots$, for the given power allocation and channel state. Two qualifications follow. First, this is an optimum *within the adopted threshold-search framework*: no non-nested subset is ever examined, so global optimality over all $2^{|\mathcal{F}|}$ subsets is not claimed —

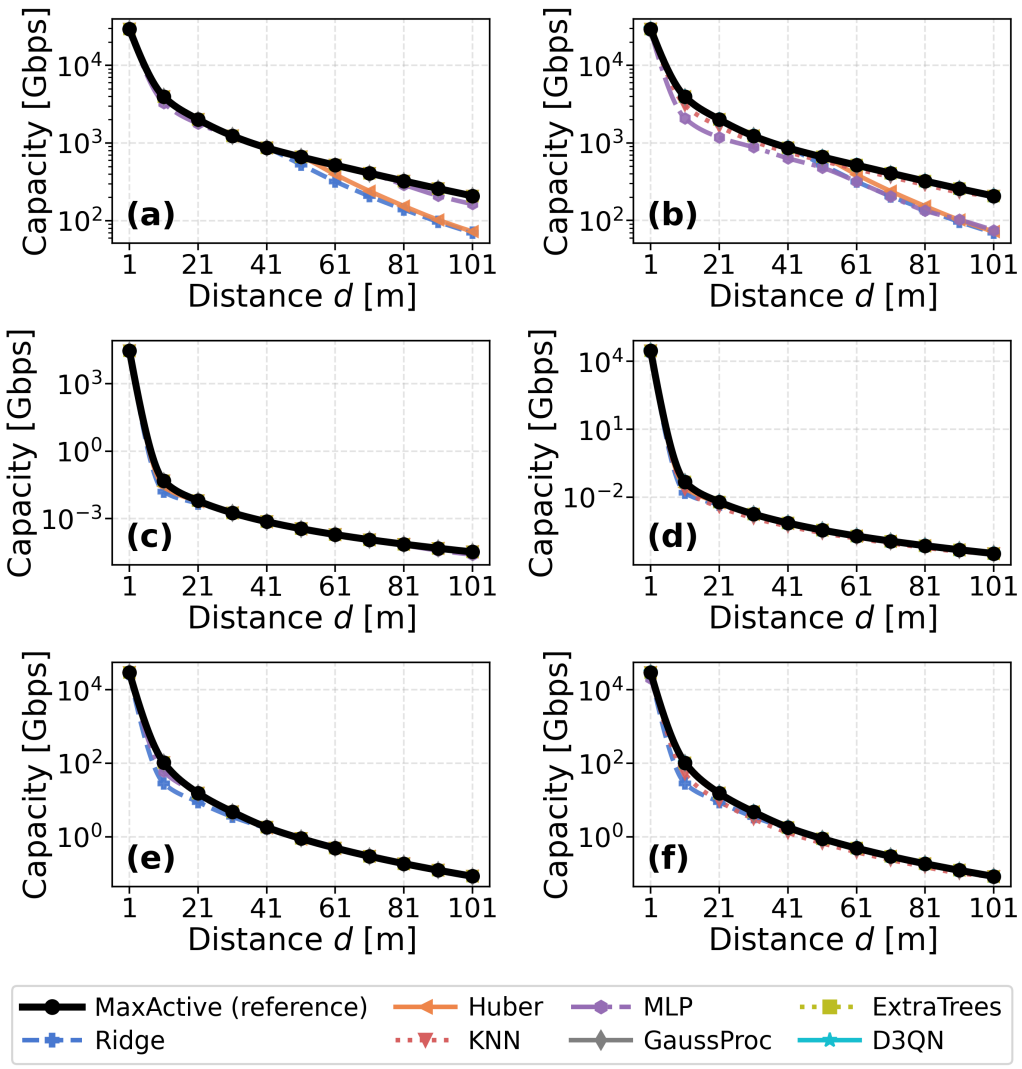



Fig. 1. Learned surrogates versus MaxActive. For legibility one representative of each model family is plotted — Ridge (regularized linear), Huber (robust), $k$-NN (instance-based), MLP (neural), GaussProc (kernel/probabilistic) and ExtraTrees (tree ensemble); the complete eighteen-model comparison is given in Table III. Rows are mobility scenarios and columns are power allocations: (a),(b) aligned; (c),(d) misaligned diagonal without beamwidth adjustment; (e),(f) misaligned diagonal with beamwidth adjustment; WF in (a),(c),(e) and EP in (b),(d),(f). Curves are models fit on all 33 samples, evaluated at the MaxActive ceiling (1) for display only, and smoothed by monotone (PCHIP) interpolation; markers denote the true computed points. The D3QN policy is shown on the EP column. Note the logarithmic capacity axis.

it would hold only under the standard argument that, at a common power level, capacity is monotone in per-channel SNR, making an optimal subset an SNR-ordered prefix. Second, selection and power allocation are optimized alternately rather than jointly, so MaxActive is optimal *conditioned* on the allocation supplied to it, not for the joint selection-and-power problem. Within these assumptions — the LBLRTM channel model, the spherical-lobe antenna model, and the alternating optimization of Section VI — MaxActive is the best decision available. It is in this restricted sense that it serves as the learning target and as the reference ceiling in (1), and "optimal" should be read with this scope throughout the paper.

### C. D3QN as a Single-Step Contextual Decision Problem

The D3QN learns a channel-selection policy from a channel-state vector. For $N$ candidate channels the

**Algorithm 1** MaxActive Channel Selection

**Given:** $\Delta f, \Delta s, z^t_{Tx}, z^t_{Rx}, d^t, f_0 \le f \le f_Q$ and $j = 1$
Find maximum SNR:
  $s^t_j = \max\left(\gamma^t(f, z^t_{Tx}, z^t_{Rx}, d^t, \Delta f)\right)$
**while** $(j \le Q)$ & $(s^t_j \ge 0)$
Find channel set $\Omega_j : \gamma^t(f, z^t_{Tx}, z^t_{Rx}, d^t, \Delta f) \ge s^t_j$
Calculate capacity:
  $C^t_{\Omega_j} = \sum_{f_i \in \Omega_j} \Delta f \, \log_2[1 + \gamma^t(f_i, z^t_{Tx}, z^t_{Rx}, d^t, \Delta f)]$
Update SNR threshold: $s^t_j \leftarrow s^t_j - \Delta s$
Move to next iteration: $j \leftarrow j + 1$
**end while**
**Return** $\Omega = \{\Omega_j \mid \max[C^t_{\Omega_j}] \; \forall \, j = 1, \ldots, Q\}$

TABLE I
KEY EXPRESSIONS OF THE THZ LINK AND THE MAXACTIVE/D3QN MODEL

| Quantity | Expression |
|---|---|
| Spreading loss (dB) | $20\log_{10} d + 20\log_{10}\frac{f}{10^6} - 27.55$ |
| Transmittance | $\tau(f,d) = e^{-k(f)d}$ |
| Absorption loss (dB) | $L_{\text{abs}} = k(f)\, d\, 10\log_{10} e$ |
| Total path loss (lin.) | $A_i = 10^{(L_{\text{spr},i}+L_{\text{abs},i})/10}$ |
| Molecular noise | $P_{N,i} = k_B T_0 (1-\tau_i)\,\Delta f_i$ |
| Antenna gain (in beam) | $4\pi/\theta_{bw}^2$ |
| Antenna gain (side lobe) | $\rho$ |
| Total link gain | $G = G_{\text{Tx}}\, G_{\text{Rx}}$ |
| Per-channel SNR | $\gamma_i = \dfrac{P_i^{\text{Tx}}\, G}{A_i\, P_{N,i}}$ |
| Capacity | $C = \sum_{i\in\Omega} \Delta f_i \log_2(1+\gamma_i)$ |
| Equal power (EP) | $P_i^{\text{Tx}} = P_T/\lvert\Omega\rvert$ |
| Water-filling (WF) | $P_i^{\text{Tx}} = \max(0, \mu - A_i P_{N,i}/G)$ |
| MaxActive selection | $\Omega = \arg\max_j C^t_{\Omega_j}$ (Alg. 1) |
| D3QN reward | $r = \ln\big((C_{\max}-C_{\min})+1\big) - \ln\big((C_{\max}-C)+1\big)$ |
| Dueling value | $Q = V(\mathbf{s}) + \big(A(\mathbf{s},a) - \overline{A}\big)$ |

state stacks two per-channel physical descriptors, the log path loss and the log absorption-noise power, $\mathbf{s} = [\log_{10} A_1, \ldots, \log_{10} A_N, \log_{10} P_{N,1}, \ldots, \log_{10} P_{N,N}]$. Given $\mathbf{s}$, the greedy action $a^\star = \arg\max_a Q(\mathbf{s}, a; \theta)$ maps to a number of selected channels $N_{\text{sel}}$; the $N_{\text{sel}}$ highest-SNR channels are activated with equal power, the SNR and capacity are recomputed, and the capacity-driven reward of Table I drives the achieved capacity $C$ toward the best achievable $C_{\max}$. The network uses the dueling decomposition [4], in which a shared trunk feeds a scalar value stream $V(\mathbf{s})$ and an advantage stream $A(\mathbf{s}, a)$, and it is trained with the double-DQN target [2], [3] under a smooth-$\ell_1$ loss.

*This is a contextual bandit, not sequential reinforcement learning.* We stress the formulation because it departs from the usual reading of a "deep $Q$-network." The problem is *single-step*: for a given geometry the channel state $\mathbf{s}$ is fixed, the agent takes one action, receives one reward, and the episode terminates. Selecting channels now does not alter the state at the next geometry, which is set by the trajectory and the physics rather than by the previous action. With no transition to influence, the horizon is one and the discount is zero, so $Q(\mathbf{s}, a)$ reduces to an estimate of the immediate reward and the model is a contextual $k$-armed bandit with $k = 600$ arms.

Two consequences follow. The mechanisms that make deep reinforcement learning fragile — bootstrapping bias, moving targets, reward delay — are absent, which is why training converges in 20 epochs across all four mobility patterns. And the close agreement with MaxActive reported later is not evidence that deep reinforcement learning solved a hard sequential control problem; it shows that a learned one-shot rule can reproduce a search whose structure is itself memoryless. We retain the dueling and double-$Q$ machinery for fidelity to [3], [4], but claim no sequential decision-making.

*Operation at an unseen geometry, and its cost.* The policy is fitted per operating context, one lightweight network for each (distance, gain) pair. This is a real restriction and we state it as such: a previously unseen geometry is not covered by an already-trained network, and obtaining a policy for it requires its own training run. That run is short in absolute terms (Section VIII-B), but it is an offline cost, so the present D3QN branch suits a trajectory envelope that can be enumerated and trained for in advance rather than an arbitrary geometry encountered in flight. The supervised branch does not share this restriction: a single fitted regressor maps the feature vector of any geometry inside the trained envelope to a capacity estimate. Two routes would lift the restriction, and both are left to future work: conditioning one network on the same geometry descriptors the regressors already use, so that a single policy spans the trajectory space; or training a bank of policies offline and selecting among them by nearest context at run time.

## V. PROBLEM DESCRIPTION

The transmitter is fixed and the receiver drone moves. Transmit power is 24 dBm, the active THz band is 0.75–4.4 THz, and both antennas use a 10° beamwidth with side-lobe level $\rho = 0.1$. Per-channel molecular absorption is taken from LBLRTM transmittance files, which fix the straight-trajectory distances to $\{1, 11, \ldots, 101\}$ m. All four trajectories are generated from the reference MATLAB model — the LBLRTM data, initial coordinates, velocities, and the hemisphere and acceleration integrators — so no value is taken from a published figure. Four mobility patterns are studied, the last three misaligned and each evaluated with and without beamwidth adjustment:

1) *Aligned:* the receiver recedes along a straight path with the beams kept aligned; capacity is highest and decays smoothly with distance.
2) *Misaligned, diagonal mobility:* the receiver moves diagonally so that the fixed 10° beams progressively misalign; capacity collapses by orders of magnitude.
3) *Misaligned, mobility over elevation:* the receiver moves on a constant-radius ($r$=20 m) hemisphere while the elevation/zenith angle $\phi$ sweeps $0° \to 180°$; the distance is constant, so the capacity change is caused purely by misalignment.
4) *Misaligned, real mobility traces:* the Tx and Rx positions are integrated from real windy-hover acceleration traces over 0–5 s with a diagonal drift, adding realistic jitter on top of the misalignment.

Table II lists representative MaxActive capacities for the aligned and diagonal scenarios. It confirms the ordering aligned > adjusted > non-adjusted over about nine decades, and the near-equality of WF and EP; the full curves are in Fig. 2. The elevation and real-trace behaviors appear, together with the ML predictions, in Fig. 3. Without adjustment, the elevation capacity stays high while the receiver is inside the half-power beam, then drops to a side-lobe plateau once $\phi$ exceeds it; adjustment widens the beam and yields a gradual decline instead. For the real traces, the diagonal drift produces a sharp collapse that the measured jitter then modulates. Because the trajectory begins at the reference initial distance of 1 m, that curve starts from the corresponding high capacity rather than from any value read off a figure.

TABLE II
MAXACTIVE CAPACITY (GBPS) VERSUS DISTANCE

| | Aligned | | Diag. w/ adj. | | Diag. no adj. | |
|---|---|---|---|---|---|---|
| $d$ (m) | WF | EP | WF | EP | WF | EP |
| 1 | 29421 | 29412 | 29421 | 29412 | 29421 | 29412 |
| 21 | 2013 | 2006 | 15.40 | 15.34 | $6.2\times10^{-3}$ | $6.2\times10^{-3}$ |
| 61 | 518.1 | 516.5 | 0.491 | 0.491 | $1.9\times10^{-4}$ | $1.9\times10^{-4}$ |
| 101 | 208.3 | 207.5 | 0.084 | 0.084 | $3.3\times10^{-5}$ | $3.3\times10^{-5}$ |

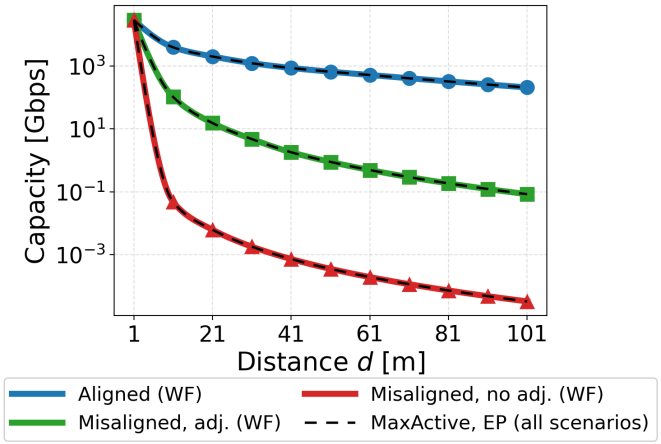



Fig. 2. MaxActive reference capacity versus distance for the aligned link and for diagonal mobility with and without beamwidth adjustment, under WF (thick, solid) and EP (thin, dashed) allocation. The three scenarios occupy disjoint decades of the logarithmic axis, spanning about nine orders of magnitude, and WF and EP nearly coincide throughout (Section VII-C). Curves are PCHIP-smoothed with markers at the true computed distances.

### A. Dataset Construction and Scalability

For the supervised study we build one sample per scenario–distance pair. The main dataset (aligned and the two diagonal scenarios) contains 33 samples; the regression target is $\log_{10}$ of the MaxActive capacity in Gbps, computed separately for WF and EP. Each sample carries a 13-dimensional physical feature vector: log-distance and its square, normalized total gain and its square, normalized beamwidth, scenario identifier, allocation flag, the mean and standard deviation of the per-sub-band log-SNR, the mean and standard deviation of the log absorption-noise, the fraction of valid sub-bands, and the normalized mean path loss. A separate 60-sample dataset ($2\times 19$ elevation points and $2\times 11$ real-trace points) is constructed identically for the additional mobility scenarios.

*Why the dataset is small, and how it scales.* The sample count is set by the reference model rather than by a sampling choice: the LBLRTM files resolve the straight-line geometry at eleven ranges, and every extra sample costs a full physics evaluation. Three properties make this regime workable rather than merely small. First, the target is a deterministic, noiseless function of the features, so there is no label noise to average out; $n$ controls only the resolution of the input space, not the variance of the labels. Second, that space is low-dimensional and physically smooth, with capacity varying monotonically in distance and misalignment, so a modest number of well-spread points pins the surface down. Third, the evaluation protocol is matched to the regime: nested cross-validation with 15 outer folds (Section VI), rather than a single split, is what allows 18 models to be ranked credibly on 33 points.

Scaling up is straightforward but changes which model to prefer. Dataset generation is embarrassingly parallel and linear in the number of operating points, at roughly 95 ms per sample, so a denser range grid, more beamwidths, additional carrier configurations, or a swarm of links would push $n$ from tens to $10^4$–$10^6$ at a cost of CPU-hours rather than new modeling. At that scale the $O(n^3)$ training and $O(n^2)$ memory of the Gaussian process become binding and the recommendation shifts to the tree ensembles (Section VIII). More data would also relax the main generalization caveat: the present models are validated as *interpolators* within the envelope of geometries they were trained on, with no claim of extrapolation beyond it. Since the reference model generates that data on demand, closing this gap is a matter of compute rather than of method.

## VI. IMPLEMENTATION

The physical layer is reproduced in Python from the reference MATLAB model: the LBLRTM transmittance is rescaled to the exact range (Beer–Lambert), and power allocation and selection are solved in an outer loop (WF or EP, then MaxActive) that iterates until the active set is stable; for the adjustment scenarios an inner search optimizes the beamwidth per step. The elevation and real trajectories are ports of the MATLAB hemisphere generator and acceleration integrator.

Because 33 (or 60) samples cannot support a single train/test split for ranking 18 models, the reported metrics use *nested* cross-validation. An outer repeated $k$-fold loop (5 folds $\times$ 3 repeats $=$ 15 folds) estimates generalization, while an inner 3-fold grid search tunes hyper-parameters on the outer training partition only. Feature standardization sits inside the pipeline, so it is fit on training folds alone and leakage is eliminated. Grid search covers regularization strength for the linear models, kernel width and penalty for SVR and kernel ridge, and depth and number of estimators for the ensembles; the Gaussian process tunes its kernel internally by marginal-likelihood maximization. Alongside RMSE and $R^2$ on the log target we report a robust *median relative error* in linear Gbps, which avoids the pathological inflation of the mean absolute percentage error at the very small far-distance capacities. Only the curves *plotted* in the figures are evaluated at the ceiling (1), each model there being refit on all data (the estimator one would deploy); every quantitative metric derives from raw, unclipped out-of-fold predictions, keeping the generalization estimate independent of the clip.

The D3QN operates on individual SNR-ranked channels (top $N = 500$, a 1000-dimensional state) at the same granularity as MaxActive-EP, with one lightweight bandit per (distance, gain) context. The dueling network maps $1000\to16\to8$ to value and advantage heads over 600 actions, trained for 20 epochs of 80 $\epsilon$-greedy actions (Adam, learning rate $10^{-3}$, weight decay $10^{-4}$).

## VII. Performance Analysis

### A. Regressors versus MaxActive

Table III reports the nested cross-validated metrics for the EP target, sorted best to worst by RMSE. The WF target was cross-validated identically; the two agree to within $0.055$ RMSE for every one of the eighteen models, and to within $0.02$ for all but the single decision tree, as the near-equality of the WF and EP capacities (Table II) implies, so only the EP column is tabulated. Three findings stand out. *The smooth non-parametric and kernel models win:* the Gaussian process is best by a wide margin ($R^2 = 0.99$, median relative error $\approx 1.9\%$), followed by ExtraTrees, SVR-RBF, MLP, and $k$-NN — they assume the smoothness the physical capacity curve actually exhibits, so they interpolate MaxActive closely and generalize across folds. *Regularized linear models are stable but biased:* Ridge, Lasso, ElasticNet, and BayesianRidge hold $R^2 \approx 0.97$ with small variance, yet their $\approx 45\%$ median relative error reveals a systematic offset, since a model affine in the features cannot capture the curvature of a capacity decaying over nine decades. *Cross-validation exposes failures a single split hides:* Huber attains the best single-split accuracy but collapses to $R^2 = 0.93 \pm 0.10$ under CV, and ordinary least squares is worse still ($R^2 = -7.98 \pm 31$), dominated by unlucky folds; with single decision trees and gradient boosting these are the failing methods. On 33 points a single split is not trustworthy; the nested-CV mean $\pm$ std is.

The last two columns of Table III quantify how far the *raw* predictions exceed the reference ceiling — what the display clip of (1) conceals, and equivalently the error a deployed surrogate would actually commit, having no ceiling to clip against. Nearly every regressor crosses the bound somewhere, so violation magnitude is a first-class selection criterion rather than a cosmetic detail. The piecewise-constant single tree and plain gradient boosting overshoot by thousands of percent because they cannot extrapolate the wide dynamic range, and a link budget dimensioned on such a prediction would be badly optimistic. GaussProc and ExtraTrees, by contrast, stay within a few percent, and the D3QN never violates the bound at all. Per-scenario, the misaligned-without-adjustment case (the worst-SNR regime) is hardest for most models, while GaussProc, SVR-RBF, kernel ridge, and $k$-NN stay within about $6\%$ across all three main scenarios, marking them as the robust choices.

Table IV breaks the out-of-fold median relative error down by scenario for the strongest models, confirming that the smooth and kernel models remain accurate even in the worst-SNR (misaligned, no adjustment) regime, whereas the tree ensembles degrade there. This per-scenario stability, not merely a low aggregate error, is what qualifies a model as a robust surrogate.

TABLE III
Nested Cross-Validated Accuracy and Physical-Ceiling Violations (EP Target, Ranked Best to Worst)

| | Generalization (nested CV) | | | Ceiling viol. | |
|---|---|---|---|---|---|
| Model | RMSE $\pm$ std | $R^2$ | Med. rel. | pts. | max |
| GaussProc | $0.100 \pm 0.23$ | 0.991 | 1.9% | 19 | 2.7% |
| ExtraTrees | $0.237 \pm 0.18$ | 0.986 | 12.1% | 1 | 3.7% |
| SVR-RBF | $0.253 \pm 0.22$ | 0.982 | 7.8% | 18 | 318% |
| MLP | $0.254 \pm 0.21$ | 0.980 | 21.1% | 20 | 226% |
| BayesianRidge | $0.331 \pm 0.14$ | 0.976 | 42.5% | 17 | 322% |
| $k$-NN | $0.338 \pm 0.29$ | 0.959 | 9.0% | 11 | 111% |
| XGBoost | $0.342 \pm 0.16$ | 0.970 | 36.9% | 14 | 38% |
| KernelRidge | $0.356 \pm 0.33$ | 0.962 | 16.0% | 17 | 223% |
| Lasso/ENet/Ridge | 0.36–0.37 | 0.97 | 45–49% | 17–19 | 232–261% |
| GradBoost | $0.412 \pm 0.21$ | 0.957 | 38.4% | 16 | 2993% |
| RF/Bag./AdaBoost | 0.44–0.45 | 0.96 | 49–62% | 9–11 | 97–246% |
| Huber | $0.504 \pm 0.37$ | 0.930 | 48.4% | 17 | 1102% |
| DecisionTree | $0.561 \pm 0.32$ | 0.931 | 48.0% | 3 | 3698% |
| LinearReg | $3.281 \pm 8.78$ | $-7.98$ | 55.8% | 15 | 405% |
| D3QN (policy) | — | — | $\sim$0% | 0 | 0% |

RMSE and $R^2$ are on the $\log_{10}$ target, the median relative error in linear Gbps; all from *raw, unclipped* out-of-fold predictions. ENet = ElasticNet, RF = RandomForest, Bag. = Bagging. "pts." counts, out of 33, the points where the prediction exceeds the reference and "max" the largest such excess — what (1) conceals.

TABLE IV
Per-Scenario Median Relative Error (EP, Out-of-Fold)

| Model | Aligned | Diag. no adj. | Diag. adj. |
|---|---|---|---|
| GaussProc | 2.0% | 1.9% | 0.8% |
| SVR-RBF | 2.6% | 5.3% | 6.9% |
| KernelRidge | 4.9% | 6.0% | 5.3% |
| $k$-NN | 1.2% | 20.1% | 0.7% |
| ExtraTrees | 2.7% | 12.2% | 24.6% |

### B. Mobility Scenarios and D3QN

Extending the benchmark to the elevation and real-trace mobility patterns, Fig. 3 overlays one representative per model family and the D3QN policy on the MaxActive curves. Fitting on all data, the strongest models reproduce these curves closely: XGBoost, GradBoost, and GaussProc reach median relative errors of $0.07\%$, $0.7\%$, and $2.8\%$ over the elevation and real scenarios (Huber, $4.4\%$, is next). Note that this is an in-sample fit, so it flatters models such as GradBoost that fail under cross-validation — which is precisely why Table III, not this fit, is the basis for the ranking. Otherwise the family ordering of Table III is visible in the figure: the kernel and instance-based representatives hug the reference while the linear ones lag. The D3QN, trained as a single-step contextual bandit at every trajectory point (Section IV-C), reproduces the MaxActive EP capacity essentially exactly on *all seven* scenario variants: the mean capacity ratio is $99.9\%$ for the aligned case and $100.0\%$ for each of the six misaligned cases — diagonal, elevation, and real-trace mobility, with and without beamwidth adjustment. The worst single operating point over all 93 evaluated geometries (33 in the main study plus 60 mobility points) still reaches $99.7\%$ of the reference, and no point exceeds it. Two points should be read together. A learned policy operating at the same individual-channel granularity as MaxActive-EP, driven by the capacity reward, recovers the reference selection rather than merely a capacity estimate, and does so robustly under elevation and real-trace mobility. But the result is bounded by construction: the policy searches the same SNR-ordered prefixes MaxActive does, so $100\%$ means *it found the same prefix*, not that it improved on the reference or solved a harder problem.

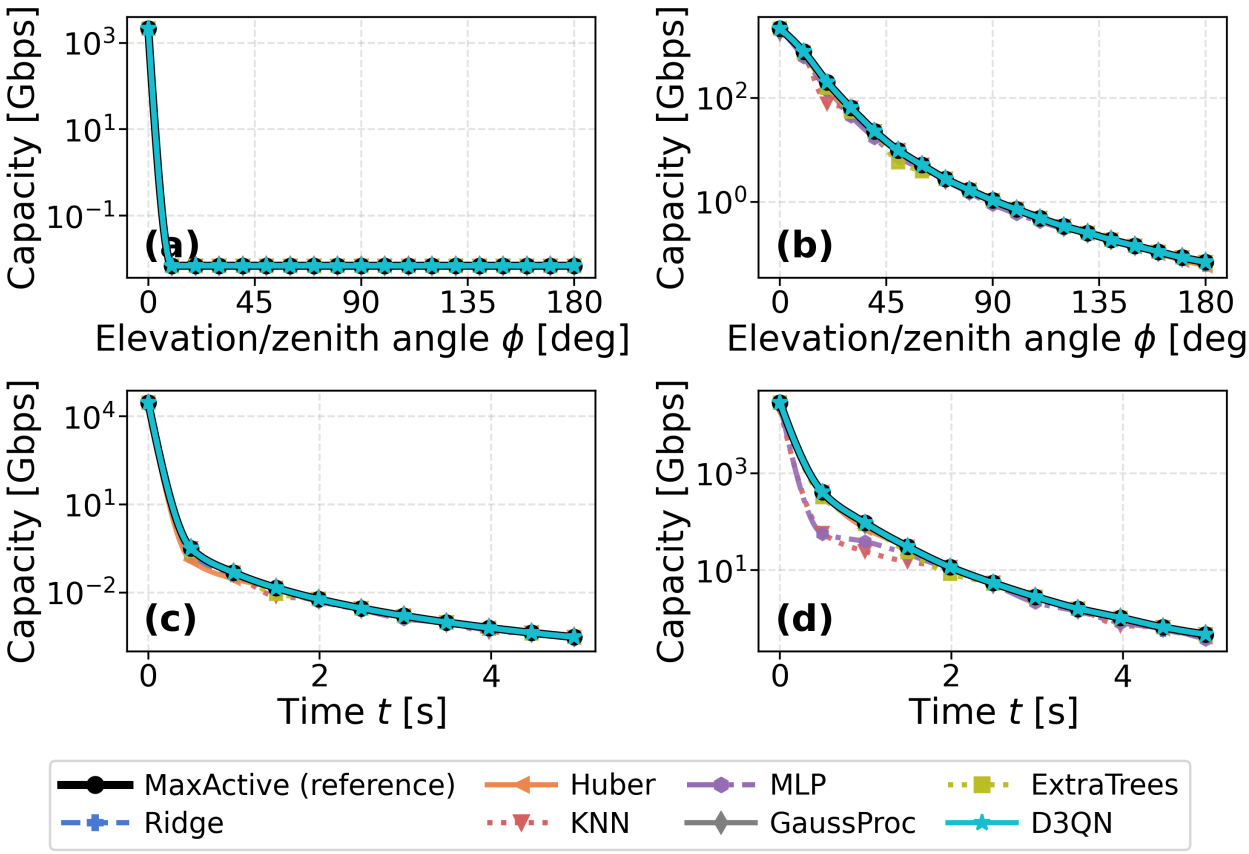



Fig. 3. Learned surrogates versus MaxActive on the two additional mobility patterns, under EP allocation. As in Fig. 1, one representative per model family is plotted and the full eighteen-model comparison is tabulated: elevation mobility (a) without and (b) with beamwidth adjustment, and real windy-hover traces (c) without and (d) with adjustment. Without adjustment the elevation capacity falls to a side-lobe plateau once $\phi$ leaves the beam, whereas adjustment gives a gradual decline; in the real traces the diagonal drift produces the collapse and the measured accelerations the residual jitter. The WF panels are omitted because WF and EP coincide to within the line width (Table II). Curves are evaluated at the ceiling for display and PCHIP-smoothed; D3QN (cyan) lies on the MaxActive curve throughout.

### C. Discussion and Limitations

Three observations round out the analysis. First, the WF and EP capacities are nearly identical across all scenarios (Table II): at THz the selected channels already have comparable, high SNR, so the water-filling gain over equal power is marginal, which justifies benchmarking the equal-power D3QN against MaxActive-EP. Second, the contrast between the value regressors and the D3QN is the most transferable finding of this study. The regressors predict a number and can exceed the physical bound. The D3QN predicts a selection, and the capacity it reports is the one its choice actually delivers, so it cannot exceed the reference — which is why it attains $100\%$ of MaxActive-EP with no clip anywhere in the pipeline. Where a surrogate must be trusted without supervision, learning the decision is the safer target than learning its value.

Third, the study has limitations that bound these claims. The datasets are small (33 and 60 points) and physically smooth, which favors smoothness-assuming models and restricts the validated regime to interpolation within the trained envelope (Section V-A). The surrogates are validated against a reference that is itself optimal only within its threshold-search family (Section IV-B), so they inherit that scope rather than approximating a globally optimal selector. The regressors predict the capacity value rather than the selection mask, so a deployment that needs the mask must use the D3QN branch. That branch is in turn realized as per-context bandits rather than a single geometry-generalizing policy, so a previously unseen geometry requires its own offline training run before the policy applies to it (Section IV-C). Finally, all timings in Section VIII come from a laptop-class CPU, so absolute latencies on an embedded airborne processor will be larger. The platform-independent quantity is the *ratio* between search and surrogate, since both slow down together on weaker hardware.

## VIII. Complexity, Runtime, and Deployment

### A. Asymptotic Cost

MaxActive has per-step cost $O(Q\,|\mathcal{F}|)$ and must be rerun whenever the geometry changes; it needs no training but incurs this search at every time step. The supervised surrogates move that cost offline. The entire nested-CV benchmark of $18$ models over both targets completes in about $30$ s on a CPU, dominated by the grid searches, and fitting the $18$ deployable models on the full dataset takes $0.5$ s. Afterwards a prediction is a single forward, kernel, or neighbor evaluation.

Table V summarizes the asymptotic training and prediction complexity of representative families ($n$ samples, $d$ features, $T$ trees, $W$ network weights, $n_{sv}$ support vectors) together with their measured cost and cross-validated accuracy, and adds the MaxActive and D3QN references for context.

### B. Measured Execution Time

Asymptotics alone do not establish a practical benefit, so every stage was measured on identical channel states, on a single core of an Intel Core i7-13620H with the band discretized into $|\mathcal{F}| = 11\,700$ active channels per context; the measured columns of Table V average the eleven aligned contexts.

The reference costs 77 ms for the threshold sweep alone and 95 ms for a complete link update with power allocation iterated to convergence. On the three geometries where both variants were timed, adding the beamwidth search raises that update from 72 ms to 91 ms, i.e., $1.3\times$ the fixed-beam cost. Against this, the surrogate path costs $0.84$ ms to turn the channel state into the 13 features and 21–43 $\mu$s to evaluate a linear, kernel, or Gaussian-process model: 0.9 ms end-to-end, an $88\times$ reduction in per-update decision cost. Where that factor comes from is worth stating precisely. The model evaluation alone is 1800–3700$\times$ faster than the search, but feature extraction now dominates the online path. The end-to-end gain therefore saturates near two orders of magnitude, and *feature extraction, not inference, is the target for further optimization*. The tree ensembles are the exception at the other end: they need 3.0–12 ms per prediction because hundreds of estimators must each be traversed, cutting their advantage to 6–20$\times$ and making them a poor per-update choice at this dataset size despite their accuracy. The D3QN is the fastest decision of all. Its state is built from the SNR ranking MaxActive computes anyway (0.55 ms, shared by both), after which a forward pass and $\arg\max$ take $0.05$ ms, three orders of magnitude below the search. It also returns the selection itself rather than a capacity estimate. Its cost is paid offline: $0.11$ s per context, about $4$ s for the 33 contexts of the main study.

### C. Memory Footprint and Airborne Deployment

For a drone, memory and predictability matter as much as speed. The deployable artifacts are small: the regularized-linear models occupy 1.2–2.2 KB, SVR-RBF $4.6$ KB, the Gaussian process 17.9 KB (its $33 \times 33$ kernel and training inputs), the MLP 70 KB, and the D3QN 21 561 parameters, i.e., $84$ KB in single precision. Every one of these fits in the on-chip memory of a microcontroller-class flight companion, and all are far below the LBLRTM tables the reference search itself must hold. Only the tree ensembles are heavy — XGBoost 522 KB and ExtraTrees $1.4$ MB — which, together with their millisecond-scale inference, make them a poor fit here despite their accuracy. Three further properties favor deployment. Inference cost is constant and data independent, a fixed number of multiply-accumulates with no search, iteration, or early exit, so the decision has a hard worst-case latency that a threshold sweep of state-dependent length cannot offer. All training is offline, so the platform performs no gradient computation and needs no floating-point accelerator. And the arithmetic compresses further: an 8-bit D3QN would occupy $21$ KB, and the linear surrogates reduce to a 13-element dot product. On these grounds the Gaussian process is the better choice where accuracy dominates (18 KB, $43$ $\mu$s) and the D3QN where the selection mask is needed (84 KB, $0.05$ ms, subject to the per-context training of Section IV-C), with the tree ensembles reserved for offline or ground-station use. These figures indicate feasibility on the memory and latency budgets of a flight companion; a full airborne demonstration remains future work.

### D. Accuracy versus Complexity, and Convergence

*Best models but higher complexity.* A clear accuracy-versus-complexity trade-off emerges, and it is the central practical conclusion of this study. The most accurate surrogate, the Gaussian process, is the most expensive to train ($O(n^3)$ time, $O(n^2)$ memory, prediction requiring the full training kernel); the next best, SVR-RBF and kernel ridge, are likewise kernel-based at $O(n^2$–$n^3)$; and the cheapest models ($O(nd^2)$ training, $O(d)$ prediction) are systematically biased ($\approx 45\%$ median error). The accuracy ranking of Table III therefore roughly tracks the cost ranking of Table V: the last few percent of accuracy is bought at a super-linear price in the number of samples.

Two regimes invert the recommendation, however. At the present scale ($n \leq 60$) the asymptotic penalty is invisible in wall-clock terms: the Gaussian process trains in $14$ ms and predicts in $43$ $\mu$s, cheaper at inference than $k$-NN or any ensemble. The most accurate model can therefore be chosen without a practical penalty, and GaussProc is the surrogate we recommend at this scale. What costs time here is the number of estimators rather than the asymptotic class, which is why the tree ensembles are the slowest predictors in the table. As $n$ grows toward the $10^4$–$10^6$ samples of Section V-A, the cubic training and quadratic memory of the kernel methods become prohibitive while the ensembles amortize their per-tree overhead, and the balance shifts to XGBoost and ExtraTrees. A reasonable guideline is therefore to prefer kernel methods while the dataset is small and tree ensembles once it is not.

Convergence is not a limiting factor for either learned method. The D3QN converges for the structural reason given in Section IV-C — each context is a single-step bandit with a deterministic reward, so there is no bootstrapping variance and no credit-assignment horizon — and 20 epochs of $80$ actions suffice to reach the reference capacity. The coupled WF/MaxActive loop inside the reference itself converges in a handful of iterations, since each MaxActive step can only refine an SNR-ordered prefix.

## IX. Conclusion

Machine-learning models can replace the online MaxActive search in mobile THz drone links, and the experiments identify which kinds, and why. Three findings carry beyond this system. First, the inductive bias that matters here is smoothness: the physical capacity surface is smooth and wide-ranging, so kernel and Gaussian-process models track the reference

TABLE V
ASYMPTOTIC AND MEASURED COST VERSUS ACCURACY OF REPRESENTATIVE MODELS

| | Asymptotic cost | | Measured (single core, $n = 33$) | | | Nested-CV accuracy (EP) | |
|---|---|---|---|---|---|---|---|
| Model (family) | Training | Prediction | Train [ms] | Predict [ms] | Memory [KB] | $R^2$ | Med. rel. err. |
| Ridge / Lasso (linear) | $O(nd^2)$ | $O(d)$ | 0.7 | 0.023 | 1.2 | 0.97 | 45% |
| $k$-NN (instance-based) | $O(1)$ (store) | $O(nd)$ | 0.5 | 0.211 | 7.6 | 0.96 | 9% |
| DecisionTree (tree) | $O(nd \log n)$ | $O(\text{depth})$ | 0.4 | 0.029 | 5.3 | 0.93 | 48% |
| XGBoost / ExtraTrees (ensemble) | $O(T\, nd \log n)$ | $O(T \,\text{depth})$ | 76–103 | 0.32–11.7 | 522–1362 | 0.97–0.99 | 12–37% |
| MLP (neural) | $O(\text{it} \cdot nW)$ | $O(W)$ | 49 | 0.031 | 70 | 0.98 | 21% |
| SVR-RBF (kernel) | $O(n^2 - n^3)$ | $O(n_{sv} d)$ | 0.8 | 0.038 | 4.6 | 0.98 | 8% |
| **GaussProc** (kernel/prob.) | $O(n^3)$, mem $O(n^2)$ | $O(n)$ mean | 14 | **0.043** | **17.9** | **0.99** | **1.9%** |
| MaxActive (reference) | — (online search) | $O(Q\|\mathcal{F}\|)$ / step | — | **76.8** | — | reference | — |
| D3QN (policy) | $O(E \cdot A)$ / context | $O(\text{forward pass})$ | 115/ctx. | 0.05 | 84 | $\approx$ ref. | $\sim$0% |

Prediction times are per operating point and exclude the 0.84 ms feature extraction shared by all regressors (Section VIII-B); the MaxActive row is the threshold sweep alone, on the same channel states. Memory is the serialized model size; for the D3QN it is 21 561 parameters in single precision.

closely while piecewise-constant learners cannot extrapolate it and overshoot the physical bound by orders of magnitude. Second, on datasets this small the evaluation protocol decides the ranking — a single split reorders the models and hides overfitting, so nested cross-validation is not a refinement but a requirement. Third, and most transferable, learning the *decision* is safer than learning its *value*: a policy that selects channels and reports the capacity its choice actually delivers inherits the reference's feasibility guarantee, whereas a value regressor can predict a rate the link cannot support.

Practically, the surrogate is limited by feature extraction rather than by inference, and its memory footprint is compatible with a microcontroller-class flight computer. The accuracy-optimal model therefore appears deployable at this scale, and only as the dataset grows by orders of magnitude does the super-linear cost of the kernel methods favor tree ensembles. Two caveats bound this. The D3QN branch is trained per operating context, so it covers a pre-enumerated trajectory envelope rather than an arbitrary unseen geometry, and all timings come from a laptop-class CPU. Future work will therefore replace the per-context bandits with a single geometry-generalizing policy, predict the selection mask rather than the capacity value, validate on embedded hardware, and extend the dataset into the regime where the complexity trade-off binds.

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